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Commentary

Machine Learning and Deep Learning in Medical Imaging: Intelligent Imaging

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ABSTRACT

Artificial intelligence (AI) in medical imaging is a potentially disruptive technology. An understanding of the principles and application of radiomics, artificial neural networks, machine learning, and deep learning is an essential foundation to weave design solutions that accommodate ethical and regulatory requirements, and to craft AI-based algorithms that enhance outcomes, quality, and efficiency. Moreover, a more holistic perspective of applications, opportunities, and challenges from a programmatic perspective contributes to ethical and sustainable implementation of AI solutions.

Keywords: Medical imaging; artificial neural network; deep learning; convolutional neural network; artificial intelligence

Introduction

The emergence of artificial intelligence (AI) in medical imaging has heralded perhaps the greatest disruptive technology since the early days of Roentgen, Becquerel, and Curie, but certainly since the days of Hounsfield and Anger. In medical imaging, the artificial neural network (ANN) is the backbone of machine learning (ML) and deep learning (DL). An ANN is an analysis algorithm composed of layers of connected nodes. The inputs may be radiomic features that have been extracted from the image files or if using a convolutional neural network (CNN) may be the images themselves. AI in medical imaging ushers in an exciting era with re-engineered and re-imagined clinical and research capabilities. An important driver of the emergence of AI in medical imaging has been the enhancement of visual recognition using AI in radiology to produce lower error rates than the human observer since 2015 [1,2]. Specific capabilities of ML in medical imaging include, without being limited to, detection and classification of lesions, automated image segmentation, data analysis, extraction of radiomic features, prioritizing reporting and study triage, and image reconstruction [1,3–5]. An ANN can also be used in parallel to conventional statistical analysis to provide deeper insights into research data [6].

Basics of ANN, ML, DL, and CNN

An ANN comprises nodes that can be in the order of hundreds to millions configured in a number of layers (depth) [7]. DL uses an ANN comprising many layers (eg, >6) and is generally regarded as a more sophisticated implementation of ML able to perform more detailed analysis, combining more data and/or able to represent higher levels of abstraction. Each node receives information from other nodes and the outputs from those nodes are weighted (Figure 1). The ANN aims to maximize correct answers compared with a reference (grounded truth) by adjusting the weightings on each node based on the error calculated on each forward propagation [1,7]. Through each iteration (epoch), the mathematical solution converges on a

The authors have an array of expertise and experience in radiomics, artificial intelligence, and neural networks. Each author is an invited speaker at the Intelligent Imaging Summit in April 2020 in Sydney; the preconference symposium for the 50th annual scientific meeting of the Australian and New Zealand Society of Nuclear Medicine (http://www.anzsnmconference.com/ANZSNM2020/).

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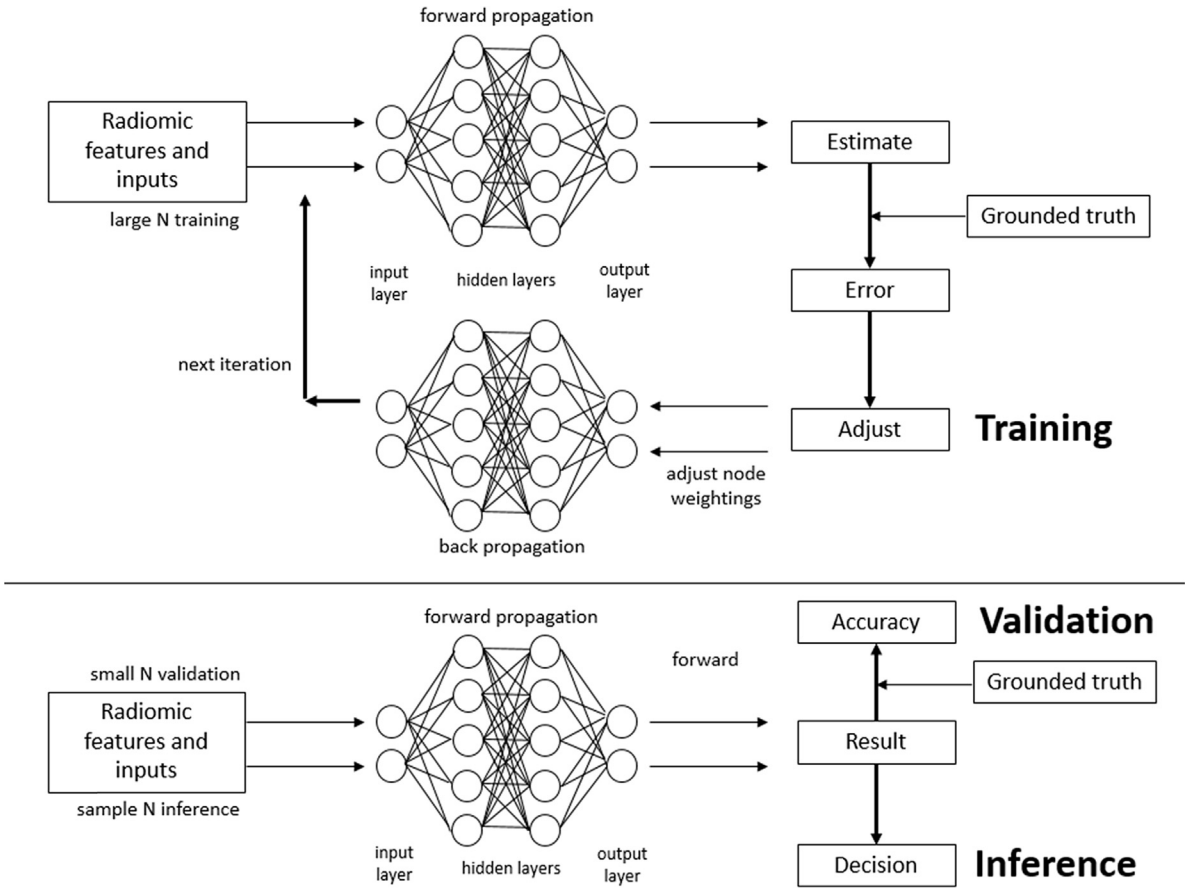

Figure 1. ANN using extracted radiomic features as inputs with a grounded truth in this supervised ANN being used for training and validation phases. After validation, the forward propagation could be used to make inferences about inputs without a grounded truth.

more accurate solution. This iterative process is not dissimilar to the process described for iterative reconstruction [8].

The training phase has the best outcomes with a large data set. With each iteration, the respective improvement in results gets marginally smaller [7]. A second, generally smaller, data set can be used to validate inferences, and this represents much of the published work at this point in time. Big data in medical imaging plays an important role in providing large, reliable training data for ML and DL algorithms to learn from. DL describes the depth or number of layers in the ANN and is generally associated with CNNs to identify and extract features directly from images [9].

Consider a simple ANN (Figure 2) with a number of input features (1 … N) and a single binary output (disease or no disease). The ANN architecture would include N scaling layer inputs, a number of hidden (perception) layers (creating depth) of multiple nodes in each hidden layer, unscaling layer, and probabilistic layer; in this case, 4 hidden layers of 3, 7, 7, and 2 nodes, respectively (Figure 2). The scaling layer contains input statistics and scales the input to a predetermined range. Each node (perceptron) in the hidden layers receives weighted (W_1, $W_{\ldots}$, W_N) numerical inputs (X_1, $X_{\ldots}$, X_N) summed and added to a bias to produce a single net input value. An activation function (A), typically either a linear or logistic (sigmoidal) activation function, then shapes the output of the node [9–13]. Each node only has 1 output but that single output becomes the input for multiple nodes in the next perception layer. The ANN has a probabilistic output function that follows an unscaling layer [11–14].

To train and optimize the ANN, a loss index is used to measure the error term (error associated with algorithm estimate) and the regularization term (size of change in outputs in response to change in inputs) [9,10,12–15]. The error and regularization terms are summed (loss index), and where the loss exceeds a predetermined value, an optimization algorithm is used to adjust the weights and bias by back-propagating the errors from the output layers toward the input layers [11,12]. The process continues, minimizing the loss index, until a predetermined value is reached (Figure 2). An epoch refers to the entire data set undergoing forward propagation (and back propagation through the ANN) once. In small data sets, this equals an iteration. In large data sets, the entire data set may not be able to be passed through the ANN in one batch. In this situation, the larger data set is broken into smaller batches. As each batch undergoes forward and back propagation, it is referred to as an iteration. Once all batches have been forward- and back-propagated, this is an epoch.

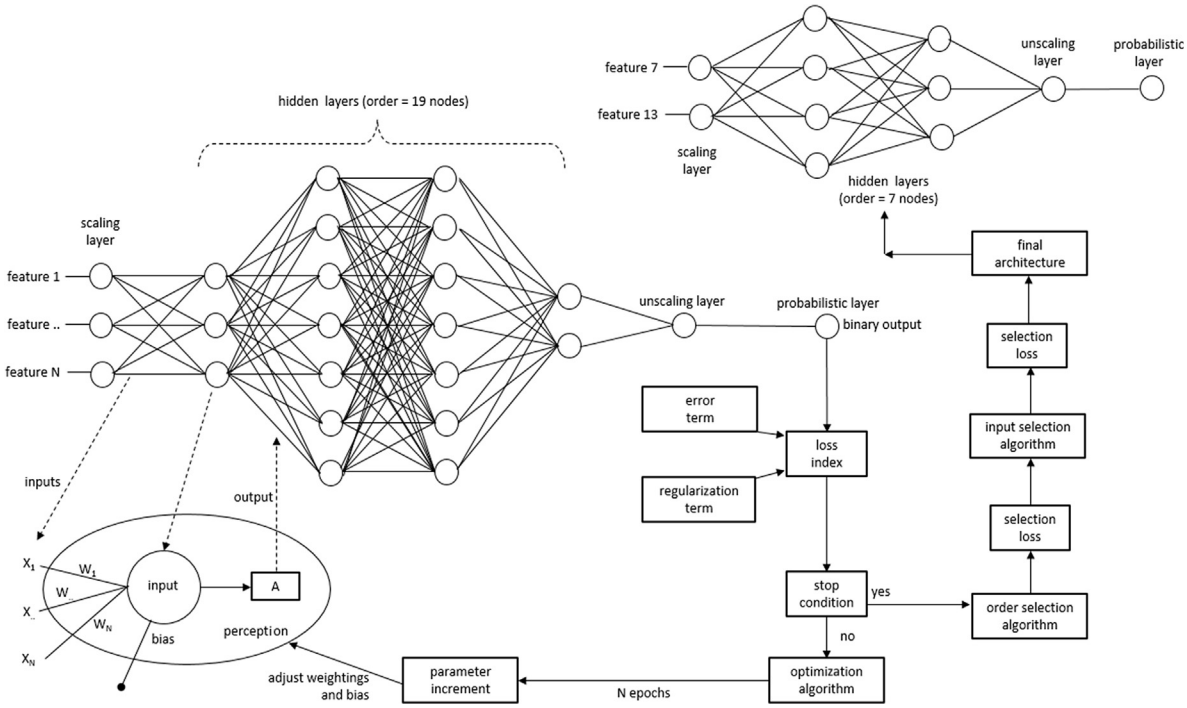

Figure 2. Overview of the anatomy of an ANN. ANN, artificial neural network.

Although the loss function estimates the error between the prediction and the grounded truth, the selection loss is a measure of the error associated with generalizability to new data [10–12]. Each contributes to optimizing the final architecture which may result in changes to the number of nodes in each hidden layers. The final architecture of the ANN considers selection loss to minimize the error associated with the order of inputs and the number of inputs [9–12]. Order selection defines the depth of the ANN or the number of nodes in hidden layers by balancing the complexity of the data with the ANN depth, avoiding overfitting or underfitting [10–12]. Input selection determines which specific inputs are necessary to be included in the ANN. Indeed, some inputs are redundant and their inclusion increases error. The input selection algorithm determines which combination and subset of inputs produce the smallest selection error [10–12].

A CNN comprises convolution and pooling layers that allow features to be extracted from the images and produce an output that is some form of classification [9] (Figure 3). Convolution extracts image features and applies a kernel (typically 3 × 3) to the input tensor (subset array of pixels) [9–11,13–16]. The kernel is superimposed over pixels in the input tensor with the distance between each successive position representing the stride [8–10,12–14]. A stride greater than 1 produces downsampling that might be best left until the pooling. The product of each input tensor pixel and the kernel are summed to produce a single numerical value in the feature map [9–11,13–16]. Multiple convolution layers are produced by applying multiple kernels to the data. The feature maps are then passed through an activation function, typically the rectified linear unit, before entering the pooling layer [9–11,13–16]. The pooling layer downsamples the feature maps to reduce computational resource requirements of the network, typically using the max pooling approach [9–11,13–16]. Max pooling creates an output equal to the maximum value within a patch of data in the feature map [9–11,13–16]. A 2 × 2 filter with a stride of 2 means that each set of 4 elements is represented as a single value equal to the maximum value (Figure 3) and can be used to select the patch that most represents the corresponding image region (eg, vertical vs. horizontal edge). Multiple sequential convolution, kernel, and pooling steps result in numerous layers of data that are transformed into a 1-directional array through a process called flattening [9,10,14].

Value of AI in Radiomic Feature Extraction and Selection

Radiomics is a term coined by Lambin et al [17] to describe the process of extracting useful imaging features from radiological data (including nuclear medicine data). The ultimate goal of radiomics is to link these features with outcomes so as to enhance precision medicine [18]. For oncology, this process involves collecting and processing the medical images, delineating the lesion of interest, extracting radiomic features, and then using this insight in tandem with conventional semantic insights to predict outcomes or response to therapy (Figure 4).

Conventionally, radiomic features determine a single scalar value to define a complete three-dimensional (3D) tumour volume; however, more recent research uses pixel-based features generating many values per feature for a 3D tumour volume [19]. These features can then be fed into a classifier to determine the features which have the strongest correlation with outcomes. An example of a useful classifier for this would be a decision tree. The classifier starts with the feature which

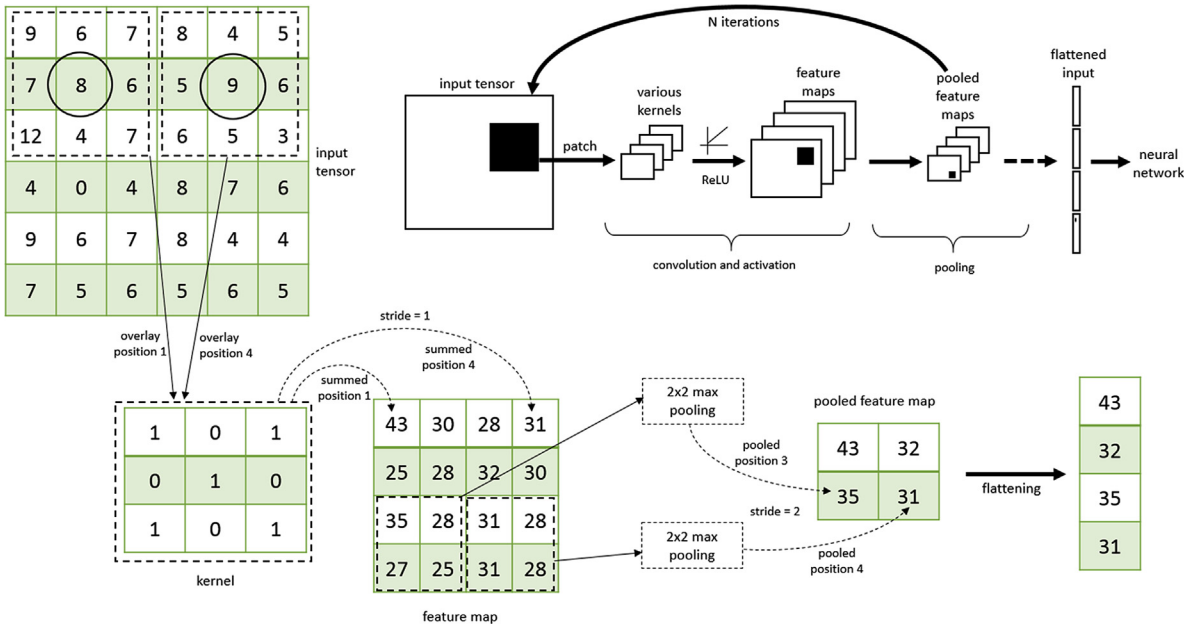

Figure 3. The CNN has a number of convolution and pooling layers before flattening and input to the neural network. Schematic representation of convolution using a 3 × 3 kernel to run sequential (in this case, successive to provide a stride of 1) 3 × 3 array of elements. The weighted sum of the kernel for the 3 × 3 input tensor creates a single representative value in the feature map. Multiple feature maps are produced by different kernels. Pooling using the max pool method and a 2 × 2 array produces pooling of the maximum count among 4 connected elements (patch) to represent those data in the pooled feature map. Consecutive blocks of 2 × 2 elements means a stride of 2. The final pooled feature map immediately before input into the neural network can then be flattened from two-dimensional data into a single dimension; this approach avoids the need for global pooling.

has the highest correlation with outcomes and then progresses down the tree using the next best feature. In this way, ML can be used to determine the most significant features and the combination of features that provides the greatest predictive capability, eliminating redundancy. Redundancy refers to individual features that have a strong correlation with other features or feature clusters. Removing redundancy in mathematical modeling reduces conflation of errors.

ANNs are data-driven and produce results that are limited by the quality of the input data. In radiology and nuclear medicine, an image or set of images may be fed into a CNN or extracted radiomic features can be used as the input for an ANN. The CNN, however, is a powerful tool in identifying and extracting its own radiomic features from the input image and link these to the outcome for improved results.

In oncology, because tumours are 3D volumes, the question arises whether it is better to simplify this 3D information by generating scalar mathematical radiomics features or whether it is better to treat the images as data and enter the raw 3D data into a CNN. Data-driven approaches such as CNNs, however, have a higher risk of overfitting to the original training data and not being generalizable to unseen data.

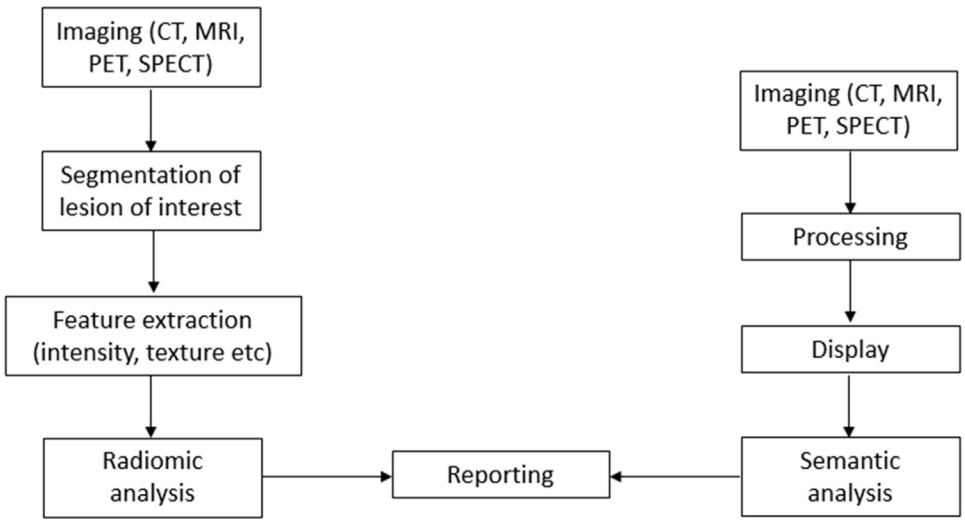

Figure 4. The radiomics workflow and integration with traditional (semantic) evaluation.

Recent developments utilizing 3D conditional generative adversarial networks contain a generator and a discriminator network in opposition [20]. These networks produce positive and negative input data from the original data, training the network for greater generalizability.

AI and Big Data, a Programmatic Perspective

The Task Ahead

As the saying goes, if AI is the engine that will drive forward the future of technology, data are the fuel on which it relies. The algorithms used to develop predictive AI models require vast amounts of data to function. Those data need to be accessible, useable, valuable, and often validated. The establishment of an AI initiative at a departmental or institutional level is a large undertaking. On the surface, it would seem that many of the components are in place. Many medical systems, from large academic programs to smaller regional networks, generate a large amount of patient-centered data on a daily basis. Those data typically exist in a state unusable for AI initiatives, residing in clinical storage systems, containing patient identifiers, unorganized and not curated. Commitment to a forward-thinking health system built around AI requires investments in the IT infrastructure to condition the data for consumption.

In the field of imaging, this may start with an image warehouse, into which imaging data are sent in a DICOM form. Ideally the data are either stripped of unique patient identifiers, or better yet, indexed according to a system which preserves linkage to clinical data sources while protecting patient confidentiality. As many will attest, this is no small undertaking, requiring a two-pronged strategy of the retroactive population of historical data, alongside the prospective population of new data.

The linkage of raw imaging data with relevant clinical information is also a challenging undertaking. Often the information sources reside in two or more separate systems, which are sometimes well-integrated, but often poorly or nonintegrated. Yet to fully realize the power of big data in the radiology/nuclear medicine realm, correlative and validating clinical information is critical.

Finally, there must be a process around data curation. AI algorithms must be trained to hone their predictive abilities, and that training needs to occur using a data set containing "truth." In the space of imaging, this may necessitate placement of bounding boxes around abnormalities in a manual fashion and in volumes large enough to support a robust training process for the AI. This curation is often manual, making use of human experts to annotate DICOM images to train the algorithms to detect and react appropriately.

The development of big data archives to support AI is, therefore, a challenge but necessary to create the environment in which AI systems and products can be introduced and can flourish. But even short of a fully dedicated initiative to develop AI, there may be investments of time, money, and infrastructure to support fully developed AI products. Some products are vendor specific and might require changes to the equipment install base to be implemented. Products which are not vendor specific might still require configuration to interact with the local environment.

The Value Question

As can be seen from the brief description mentioned previously, the commitment of a department or institution to AI initiatives brings with it a requirement of substantial resources. There will be investments needed for hardware, software, human expertise (both computational and medical), and the systems to support it all. Although from an academic and intellectual standpoint, the investments in AI may align with the overall goals of a medical system, the question remains whether the investment can be justified from a programmatic perspective.

This issue can be viewed from a generic perspective on the implementation of AI in a commercial setting; that is, via an assessment of the impact of AI on the status quo. The performance of the practice of "clinical radiology," to use the term broadly, is highly complex. The overall environment from the perspective of patient experience can be viewed as a cycle (Figure 5). To use the hypothetical example of a diagnostic imaging procedure, the cycle would originate with the identification of a need for the diagnostic imaging procedure. On identification of the need, an order may be placed, after which action is taken in response to the order: the study is scheduled, the patient contacted, and the examination protocoled. At the time of the encounter, the patient is screened and prepared, and the procedure performed. For diagnostic studies, the images are then placed in a queue to be interpreted. A radiologist/nuclear medicine physician views the study and issues a dictation, and the results are sent to the originating provider. Finally, action is taken based on the imaging.

In considering this cyclical view of the diagnostic imaging experience, one can immediately identify steps in the process in which AI technology could have an impact. For example, the initial step of deciding an imaging procedure is necessary and is essentially one of the predictions and probabilities. In addition to deciding that an imaging procedure would help address clinical uncertainty, the decision needs to be made as to which imaging procedure would be best. Although simple to write into a sentence, this concept has been the focus of many initiatives, focus groups, and white papers. The concept of appropriate use criteria is based on the need to provide guidance as to the "best" procedure for a given clinical circumstance, encompassing factors such as diagnostic power, cost, radiation exposure, and others. Even still, appropriate use criteria are only meant to be applied in broad terms, with exceptions needed and granted for particular circumstances.

Of course, a great deal of attention has focused on the application of AI toward image interpretation. Products currently exist, which offer image interpretation. Currently, these products are typically designed around a specific,

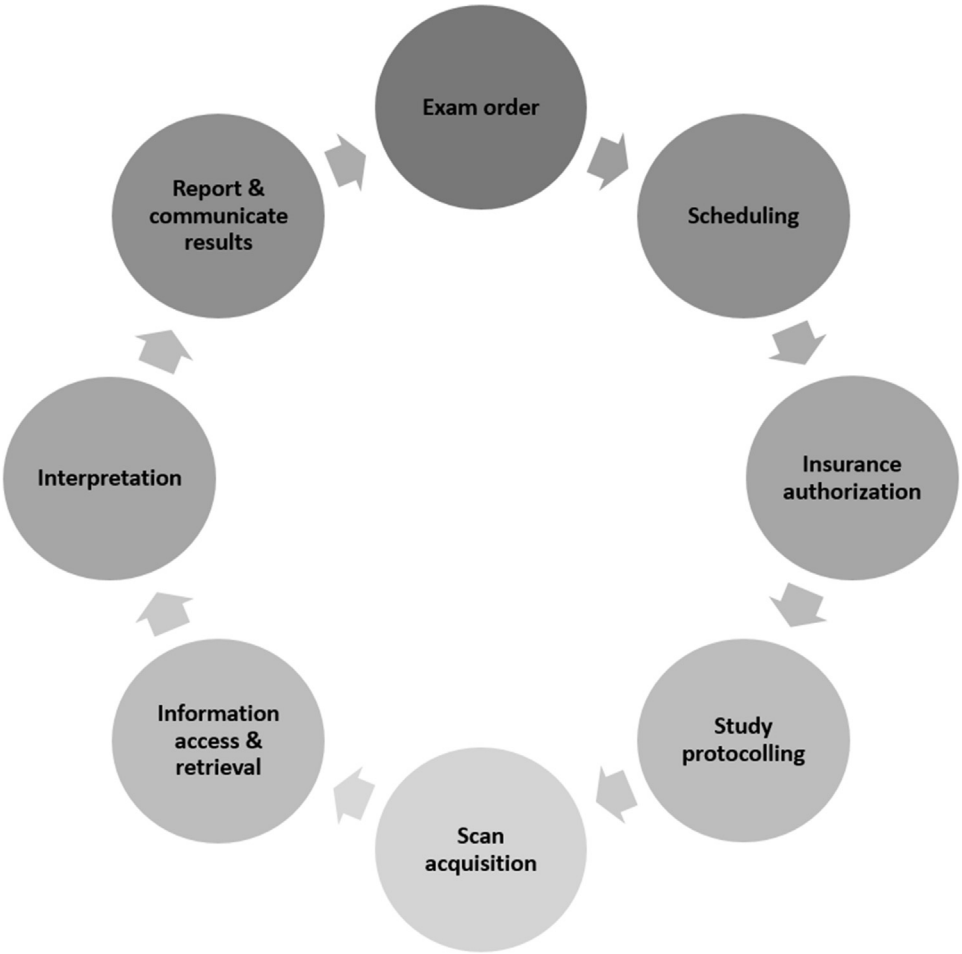

Figure 5. Representation of the examination cycle.

clinically impactful diagnostic finding, such as the identification of a pneumothorax on a chest radiograph, or the identification of suspicious calcifications on a mammogram. The potential benefit of AI in this realm depends on where in the process the AI algorithm is applied. A scanner-side algorithm can alert the radiologist/nuclear medicine physician to possible critical findings, shortening the time to notification and potentially improving patient outcome. A PACS-side (picture archiving and communication system) algorithm could improve the accuracy of diagnosis and decrease the potential for missed findings.

Each step in the cycle can be similarly evaluated, with the identification of the ways in which the power of prediction (the hallmark of AI) can provide benefit. The task from a programmatic perspective is then to assess whether the benefit justifies the cost. In the current state of technology, the application of AI technology is often fairly incremental, rather than transformative. A hypothetical algorithm may have a function that results in the identification of an important finding with a 5% higher sensitivity (without a corresponding loss of specificity) than an average human radiologist/nuclear medicine physician. It may highlight the presence of a critical result 10 minutes sooner than would occur during the course of routine operations.

Naturally, it is impossible to fully flesh out such hypothetical scenarios, with all the variables that come into play. For example, the ''value'' of a 5% increase in examination performance is linked to the impact on outcome for that particular feature. In the case of occult breast cancer, a 5% increase in the true positivity of disease detection might lead to the more diagnosis of breast cancer in a group of patients a year or earlier than would otherwise have been diagnosed. In some of these patients, quicker diagnosis could make the difference between detection of disease at an early stage rather than a late stage, with corresponding improvement in survival. Many uncertainties are at play, of course. In some patients, a delay of 1 year may not change stage, prognosis, or therapeutic options. Human skills vary, so an algorithm that may have superior performance compared to one radiologist/nuclear medicine physician may actually be inferior to another. Nonetheless, when taken in an aggregate, the potential advantages of a particular AI product can be weighed inside a particular program to determine whether those advantages justify the cost and efforts of implementation. For many of the early AI products, the balance was not tipped in the favor of implementation, but as performance improves and costs decrease, the value equation is rapidly tipping into favorability.

A Perspective on the Future

The impact of AI on the practice of radiology and nuclear medicine is a moving target, with continual advances in technology leading to improvement in performance and expansion of applications. This will require constant re-evaluation of the value question, to determine when investment is justified. It is certain that AI will play a role in the future of medicine, although the pace and pathway toward implementation will vary site by site.

To try and capture a sense of what truly transformative AI would look like in the field of radiology and nuclear medicine, one needs to imagine what could be achievable were the AI to perform perfectly, in other words, achieve 100% accuracy in its predictions. Again, the impact of this will vary depending on the scenario. What if there were a flawless AI algorithm that could evaluate every pixel of a contrast-enhanced CT of the chest, abdomen, and pelvis (in ways beyond what is available to the human eye including texture analysis, and so on) and could correlate with all available demographic and clinical data ingested from the electronic medical record? And what if through these processes, the algorithm could predict with 100% accuracy such information as the presence of occult malignancy, the risk of coronary event, and the need for dietary modification?

It is easy to make justifications as to why these far-forward possibilities would never come to fruition but only time will tell what may come about long term. It would be a mistake to dismiss things out of hand merely because they seem unlikely with the current state of technology. And it may be that there are curtains of biological uncertainty which cannot be pierced by even the most advanced technology. The power and value of AI technology, however, will absolutely progress with time.

At the current state of technology, there are serious considerations at a programmatic level as to whether to adopt current generations of AI technology. In some environments, the answer may be yes. In others, the answer may be no for now. It is almost certain that in time, AI will permeate the practice of radiology and nuclear medicine at all points in the examination cycle, sometimes visibly in the interactions with patients and physicians, and sometimes invisibly, built into the underlying systems. The pace at which that transition occurs will depend on the state of the technology and the value it brings to patient care. Programs and institutions will need to constantly evaluate the impact of these changes to make local decisions on investment and adoption.

DL in Diagnosis and Therapy

The most commonly quoted applications of DL in medical imaging applications have been for object detection (eg, location of a lesion), object segmentation (eg, lesion contouring), and object classification (eg, malignant vs. benign lesion) [21]. These operations are often performed in cascade and lend themselves to different medical applications. Object detection, for example, has been used extensively in computer-aided diagnosis of mammograms to point the reader to potential tumours and likewise in lung and liver CT scans. Object segmentation has been utilized in automated radiation treatment planning for delineating tumours and organs as target and dose-sparing regions, respectively. Finally, the entire field of radiomics is an exercise in lesion classification (relying on prior lesion detection/segmentation).

These applications, to date, have been application specific, without general intelligence. Hence, AI for segmenting the spleen is poorly able to segment a liver without extensive training. Many other, albeit more obscure, applications are being explored:

- Triaging—gestalt classification of images into normal versus abnormal anatomy and degree of severity for triaging urgent cases for radiological interpretation.
- Similar images—finding of previously encountered cases with similar findings to assist learning and to aid in interpreting rare and/or subtle cases.
- Image enhancement—noise reduction of medical image data (before or after image reconstruction) to enhance image quality and reduce radiation dose.
- Image reconstruction—replacing iterative image reconstruction methods with a direct transformation from sinogram space to image space.
- Attenuation correction from MR—estimation of CT and attenuation correction maps from MR data in PET/MR.
- Multimodal image coregistration including nonrigid image warping to account for patient positioning differences.
- Change detection and trending—detection of changes between baseline and follow-up studies for disease progression or response to therapy.
- Image acquisition optimization—AI for guiding technologists for optimal patient positioning, field-of-view delineation, and ultrasound probe positioning.
- Quality assurance—monitoring machine quality performance data to detect abnormal performance and predict maintenance actions.

Integrating DL into Clinical Workflow

Currently, applications are being developed to guide and assist human observers and must, therefore, include means to communicate complex information through a machine-human interface. Such an interface may be trivial for indicating the presence of a lesion and boundaries of an organ, enabling human oversight to detect erroneous results. Other applications such as triaging or image enhancement may not lend themselves to convenient human oversight and, therefore, must be clearly demonstrated to be robust before they may be used blindly in clinic.

The method of projecting AI findings to the radiologist/nuclear medicine physician also deserves careful consideration. Immediate presentation of AI findings can bias the human reader and lead to over compliance. Even delayed

presentation of results may lead the radiologist/nuclear medicine physician to simply wait for the computer-generated results. Presenting information in definitive terms (eg, benign/malignant or positive/negative) obfuscates the uncertainty around classifying the finding into one group or another. Probabilistic information, while communicating better information, may confuse interpretation rendering the AI application unhelpful.

One potential benefit of integrated user and machine interfaces is that human observers may be able to gain insight from the machine-observer as to how it was able to detect a medical condition that the human has made. Another potential is for the human to communicate feedback to the machine to enhance its learning from experience. Although this approach is encouraging for training AI for better performance and new application, this creates a regulatory challenge. What if the human is a poor reader and trains the machine to underperform? How do you regulate software versions as the machine evolves? Perhaps the solution is to transfer data to the software manufacturer where data can be cleaned and training can be supervised, but then who owns the data? A potential solution to these conundrums may be that as machines become smarter, the need for a human-machine interface may be simplified to only display results.

AI Application in Medical Imaging

AI and Design Thinking

The concept of design thinking has impacted technology and science for decades. Now more than ever, however, this concept is relevant and critical to successful innovation and implementation of AI technology in medical imaging. In the design thinking process, particularly when applied to medical innovation, one is encouraged to approach technologic challenges from a human-centered perspective. Indeed, there are many AI possibilities to explore in medical imaging, but the true impactful work begins when the scientific development process is centered around the questions "how can the patient experience be improved, and how can we design a technology to achieve this?." This is a question not easily asked by a single person with a single perspective. Successful design thinking relies on the synergy of a diversity of opinions. Although radiologists and nuclear medicine physicians can and must be at the helm in the process of AI development in medical imaging, the design thinking process should encompass the perspective of the varied stakeholders, referrers, medical radiation technologists, administrators, industry and, most importantly, including the patients. Radiologist and nuclear medicine physician oversight in research and development of AI in medical imaging is essential but must include collaboration with key stakeholders [7].

Data Usage and Development

When approaching the process of designing and implementing new AI applications, it is important to first understand the central problem around which the AI is to be designed. Thrall et al [5] nicely outline the difference between circumstantial and intrinsic challenges. Circumstantial challenges revolve around the behaviours of humans and society. Intrinsic challenges in AI design reside in how we are able to use science and build technology to create innovative solutions. Another challenge often ignored is that of communication and understanding. Learning to speak a common language and have a shared understanding between physicians, scientists, technologists, industry leaders, and patients is critical to achieving a shared vision [1]. Without diverse input from all stakeholders at the beginning of the design process, AI applications run the risk of developing fatal flaws. For example, an innovative scientific process may not be ethical in its application to patient care, or it may not be able to operate in accordance with laws and regulations [22]. Additional potential fatal flaws occur when the data are homogeneous and not from a diverse sample that accurately reflects the heterogeneity of the patient population. The developed application would, therefore, be unable to be successfully used in the target population [22]. It is essential to have both diversity in thought and diversity in data when approaching the initial development process. True collaboration among the diverse technology industry, physicians, and scientists ultimately relies on mutual understanding, realism from business and technology worlds, and optimism from the medical imaging professionals [23].

Implementation of Ideas

The concept of moving from bench to bedside has been pressing and relevant since some of the first medical imaging discoveries. The process of moving AI from the computer sciences laboratory to the forefront of patient care presents similar challenges to those faced by the original imaging pioneers. To successfully implement AI into the practical patient care experience, we need both clinical minds empowered to be able to understand and interface with this new technology, and we need technology that is able to evolve and adapt to the changing needs associated with clinical use. The infrastructure of understanding must be built during the clinical training process. Basics of AI, ML, and imaging informatics should be part of the basic medical school and residency curriculum and should be weaved through the training of physicists and technologists. Through ensuring that basic data science curriculum builds a foundation for the next generation of clinical care specialists, we build a body of health care professionals empowered to adopt and affectively use this new technology.

Perhaps equally as important as training is the purposeful ingraining of ability for successful implementation of AI solutions in order for them to evolve and change within the clinical context of application. Larson and Boland [24] provide an example in considering a new AI application for detection of urgent findings in a chest radiograph. If there is no built-in method to assess the performance of this solution, to gather data on efficiency and accuracy, then the ability to perform quality control measurements and improve performance over time becomes lost. Furthermore, quality tools and

algorithms must be paired with performance measures, monitoring, feedback, and accountability for success [24]. To successfully implement new AI solutions into clinical care, and ensure longevity of their use, users and clinical care providers must have a working understanding of the new technology and the technology itself must be well equipped to adapt and grow with the rapidly advancing field of medical imaging.

Regulation of Technology

Although the medical world has been intently focused on developing and applying new technologies, a whole new myriad of legal and regulatory challenges began to percolate. Two of the most pressing issues have quickly become reimbursement and regulation. No doubt payment for AI-related services will be approached quite differently depending on where in the global market the services are to be delivered. Whether a fee for service model, alternative payment model, or universal health care, the concept of AI applications as a potentially billable service disrupts current reimbursement plans. Many believe, however, that the implementation of AI-related services should be an expected standard of care tool that is inherently applied in the process of delivering care. Rather than a discrete billable encounter, AI applications in the medical care experience are likely to be seamlessly integrated into the traditional billable patient encounters. Indeed, an advantage of AI would be to reduce costs of providing services, obviating the need for additional billing to recover costs. Shoppe [25] concludes that AI in medical imaging will be treated as nonreimbursable business expenses.

As the implementation of AI use in medical care accelerates, the need for regulation and assurance of quality has become a pressing issue. In the U.S. market, the Food and Drug Administration has formally approved a small (but growing) number of AI-based medical imaging devices. The Food and Drug Administration's approach to evaluating and approving potential new technology has stayed closely rooted in its five "Culture of Quality and Organizational Excellence" principles; product quality, patient safety, clinical responsibility, cybersecurity responsibility, and proactive culture [26]. European regulations have followed a similar risk-based approach in evaluating new technology. In general, the responsibility of risk assessment has been put largely in the hands of the technology development companies or in independent certification bodies [22].

Ethics in AI

Perhaps the most contentious topic to navigate in AI application is the ethical questions that arise when using human data to develop human-targeted applications. Jalal et al [27] poignantly reflect on the fundamental ethical challenges of AI and conclude that looking past the technological advancements of AI, AI confronts the basic challenges associated with autonomy, beneficence, justice, and respect for knowledge. In every aspect, from data curation, design and development, clinical application, reimbursement, and quality control, ethical questions arise that must be adequately discussed and addressed on a global scale. One key concept that arises with initial handling of data is the difference between data privacy and confidentiality. Privacy is the right of an individual to maintain control over his or her own personal information. Confidentiality refers more to the responsibility of the physician or scientist entrusted with that data to maintain that person's privacy [28].

The very basic concept of who owns patient data and what is and is not allowed to be done with that data is fervently debated. Kohli and Geis [29] define the five key aspects of ethical data handling in AI as informed consent, privacy and data protection, ownership, objectivity, and inequity related to those that either have or lack the resources to process the data. These ethical challenges weave into the concept of design thinking, with development and implementation being a patient-centered process. This approach is key to ensuring our methods are always driven toward achieving the best possible outcome in the best interest of the patients we serve.

As AI is increasingly more common in clinical use, the safeguarding and safe handling of patient data becomes paramount. The physician/patient relationship has traditionally been founded on trust and now more than ever that basic fundamental trust must not become challenged or broken. What happens however when a breach does occur or an error occurs in the AI applications? What if the error results in an unfortunate clinical outcome? A number of new legal conundrums rise in assigning liability [30]. There may need to be shared liability between the technology and the clinical provider overseeing implementation of that technology. Perhaps, the degree of liability is dependent on the level of autonomy of the application. While a clear precedent is yet to be established, this challenge is one that will need to be carefully addressed.

The Patient Experience

The simplest and most effective way to navigate understanding the patient experience is to directly involve patients in the design, implementation, and decision-making process. Haan et al [31] directly asked patients about their perceptions and experiences, and unveiled six key concepts were central to how they framed their understanding of AI:

1. Proof of technology,
2. Procedural knowledge,
3. Competence,
4. Efficiency,
5. Personal interaction,
6. Accountability.

Fundamentally, patients want to have a basic understanding of the technology, how it will be applied to their own care, know that the application of this technology is safe, be ensured that the application results in increased efficiency and quality of care, know that this application ultimately enables deeper personal interactions with the care providers, and trust that both the provider and developers of the technology are accountable for the outcome.

The most profound outcome in AI applications with regard to the patient experience is enabling deeper, more meaningful interactions between the patient and the provider. Allowing automation of certain tasks can allow the medical imaging team members to spend more time interacting with the patient in a more meaningful and impactful manner and play a more central role in the patient's care team [32]. Ultimately, this shift allows for not only tremendous improvement in the patient experience, but also an improved sense of satisfaction and a renewed sense of value for the medical professional. This certainly applies not only the medical imaging community; the paradigm shift created be applications of AI in medicine has the potential to inspire a culture shift where emphasis is increased on the value of the interaction between the patient and the provider.

> "Freeing up physicians' cognitive and emotional space to interact fully with patients, rare in the elecronic health records era, can restore and enhance the patient experience." [33].

Since the initial images of the human body were acquired, the medical imaging community has a tradition of being shepherds of new information and applications of new technology. Through developing technology rooted in design thinking, applying solutions founded in a diversity of opinions, carefully and empathetically navigating regulatory and ethical challenges, and above all keeping the patient at the center of the process, the growth the AI uses in the clinical space has tremendous potential to transform the global culture of medicine.

Conclusion

AI, ML, and DL have penetrated medical imaging with minimal actual disruption. The emergence of ANN and CNN on the medical imaging landscape has the potential to enhance the overall ecosystem bringing with it diversity and sustainability. Understanding the principles and applications of AI, ML, and DL in medical imaging will facilitate assimilation and expedite advantages to practice. Nonetheless, it remains prudent to critically evaluate evidence of capability to avoid buying into hype or hysteria. Successful implementation of AI in medical imaging requires advanced technology combined with the ability to navigate ethical and regulatory challenges in a patient-centered design approach.

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